

DAY 16: List Comprehension

→ List comprehension is the short form of syntax to create a new list

syntax: → [expression loop condition]

syntax2: → [expression condition else loop]

Eg:

```
old=[1,2,3,4,6,7,8,9]
```

```
new=[i for i in old]
```

```
print(new)
```

```
old=[1,2,3,4,6,7,8,9]
```

```
new=[i for i in old if i%2==0]
```

```
print(new)
```

nested comprehension

→ using list comprehension generating list inside list

Eg:

```
any=[[i*j for i in range(1,5)] for j in range(1,5)]
```

```
print(any)
```

```
of =[[1,2,3],
```

```
[4,5,6],
```

```
[7,8,9]]
```

```
num=[num for row in of for num in row]
```

```
print(num)
```

generator

→ a generator is a special function which generate one value at a time...

Eg:

```
def data():
```

```
    for j in range(1,10):
```

```
        yield j
```

```
    j=data()
```

```
    print(next(j))
```

```
    print(next(j))
```

